Day 7 content is complete. then provide day 8content

Day 8 — IPv4 Addressing & Subnetting

Today we'll go deeper into **IPv4 addressing and subnetting**. This is one of the most important networking interview topics because it connects directly to the **Layer 3 routing concepts** you learned in Days 2 and 7.

🎯 Day 8 Goal

By the end of today, you should be able to explain:

```
IPv4 Address
       ↓
Network Part + Host Part
       ↓
Subnet Mask
       ↓
CIDR Prefix
       ↓
Network Address
       ↓
Broadcast Address
       ↓
Usable Host Range
       ↓
Routing Decision
```

---

1. What is an IPv4 Address?

An IPv4 address is a **32-bit address**.

Example:

```
192.168.1.10
```

It contains four octets:

```
192     .     168     .     1     .     10
 |             |             |           |
8 bits        8 bits        8 bits      8 bits

        Total = 32 bits
```

Therefore:

```
8 + 8 + 8 + 8 = 32 bits
```

---

2. Convert IPv4 to Binary

Example:

```
192.168.1.10
```

Each decimal number represents 8 bits.

```
192 = 11000000
168 = 10101000
1   = 00000001
10  = 00001010
```

So:

```
192.168.1.10

11000000.10101000.00000001.00001010
```

This is the actual 32-bit representation used by the networking system.

---

3. Why Do We Need a Subnet Mask?

Suppose:

```
IP = 192.168.1.10
```

Just knowing the IP isn't enough to determine which part represents:

```
Network
```

and which part represents:

```
Host
```

The **subnet mask** tells us where the network portion ends.

Example:

```
IP:
192.168.1.10

Mask:
255.255.255.0
```

Binary:

```
IP:
11000000.10101000.00000001.00001010

Mask:
11111111.11111111.11111111.00000000
```

The mask has:

```
24 ones + 8 zeros
```

Therefore:

```
Network = first 24 bits
Host    = last 8 bits
```

---

4. CIDR Notation

Instead of writing:

```
192.168.1.10
255.255.255.0
```

we commonly write:

```
192.168.1.10/24
```

The `/24` means:

> The first 24 bits are the network prefix.

Visual:

```
192.168.1.10/24

<---------24 bits---------><--8 bits-->
      NETWORK                  HOST
```

5. Common CIDR Prefixes

You should become comfortable with these:

CIDR	Subnet Mask	Host Bits
`/8`	255.0.0.0	24
`/16`	255.255.0.0	16
`/24`	255.255.255.0	8
`/25`	255.255.255.128	7
`/26`	255.255.255.192	6
`/27`	255.255.255.224	5
`/28`	255.255.255.240	4
`/29`	255.255.255.248	3
`/30`	255.255.255.252	2

For interviews, **/24 through /30** are particularly important.

---

6. Network Address

Let's take:

```
192.168.1.10/24
```

The network portion is:

```
192.168.1
```

The host portion is:

```
10
```

Therefore the network address is:

```
192.168.1.0
```

So:

```
Network:
192.168.1.0/24
```

---

7. Broadcast Address

For:

```
192.168.1.0/24
```

the host portion has 8 bits.

The broadcast address sets all host bits to `1`.

Therefore:

```
192.168.1.255
```

So:

```
Network Address = 192.168.1.0
Broadcast        = 192.168.1.255
```

---

8. Usable Host Addresses

For:

```
192.168.1.0/24
```

the addresses are:

```
192.168.1.0
        ↓
Network address

192.168.1.1
        ↓
First usable host

...
```

```
192.168.1.254
        ↓
Last usable host

192.168.1.255
        ↓
Broadcast
```

Therefore the traditional usable host range is:

```
192.168.1.1
      ↓
192.168.1.254
```

That's:

```
254 usable addresses
```

---

9. Why 254 and Not 256?

A `/24` has:

```
32 - 24 = 8 host bits
```

Therefore:

```
2^8 = 256 total addresses
```

But traditionally:

```
1 address → Network
1 address → Broadcast
```

So:

```
256 - 2 = 254 usable host addresses
```

Formula

For ordinary IPv4 subnets:

```
Usable hosts = 2^(host bits) - 2
```

10. Example: /25

Now let's make it slightly harder.

```
192.168.1.0/25
```

`/25` means:

```
25 network bits
7 host bits
```

Therefore:

```
2^7 = 128 addresses
```

Traditional usable hosts:

```
128 - 2 = 126
```

Subnet mask:

```
255.255.255.128
```

---

11. /25 Creates Two Subnets

A `/24` network:

```
192.168.1.0/24
```

can be divided into two `/25` networks.

```
192.168.1.0/25
```

and

```
192.168.1.128/25
```

Visual:

```
192.168.1.0/24
|
├───────────────┐
```

```
|                |
▼                ▼
/25              /25

192.168.1.0      192.168.1.128
      |                  |
      ▼                  ▼
.0 - .127           .128 - .255
```

---

12. First Subnet

```
192.168.1.0/25
```

Range:

```
192.168.1.0
      ↓
192.168.1.127
```

Therefore:

```
Network       = 192.168.1.0
First Host    = 192.168.1.1
Last Host     = 192.168.1.126
Broadcast     = 192.168.1.127
```

Usable:

```
126 hosts
```

---

13. Second Subnet

```
192.168.1.128/25
```

Range:

```
192.168.1.128
      ↓
192.168.1.255
```

Therefore:

```
Network       = 192.168.1.128
First Host    = 192.168.1.129
Last Host     = 192.168.1.254
Broadcast     = 192.168.1.255
```

---

14. /26

Now divide the `/24` into `/26`.

```
192.168.1.0/24
        ↓
       /26
```

`/26` gives:

```
32 - 26 = 6 host bits
```

Therefore:

```
2^6 = 64 addresses/subnet
```

Traditional usable:

```
64 - 2 = 62 hosts
```

---

15. /26 Creates Four Subnets

```
192.168.1.0/26
192.168.1.64/26
192.168.1.128/26
192.168.1.192/26
```

Visual:

```
192.168.1.0/24

├── 0   - 63
├── 64  - 127
├── 128 - 191
└── 192 - 255
```

Each subnet:

```
64 addresses
62 usable hosts
```

---

16. The "Block Size" Trick

This is extremely useful in interviews.

For:

```
255.255.255.192
```

look at the last octet:

```
256 - 192 = 64
```

Therefore the subnet boundaries occur every:

```
64
```

So:

```
0
64
128
192
```

This gives:

```
0   → 63
64  → 127
128 → 191
192 → 255
```

---

17. Example Interview Question

Question:

What subnet does:

```
192.168.1.70/26
```

belong to?

First:

```
/26
```

Mask:

```
255.255.255.192
```

Block size:

```
256 - 192 = 64
```

Subnet boundaries:

```
0
64
128
192
```

`70` falls inside:

```
64 - 127
```

Therefore:

```
Network = 192.168.1.64
```

Broadcast:

```
192.168.1.127
```

Usable:

```
192.168.1.65
      ↓
192.168.1.126
```

---

18. Another Example

Find the network and broadcast for:

```
192.168.10.150/27
```

`/27`:

```
Mask = 255.255.255.224
```

Block size:

```
256 - 224 = 32
```

Subnet boundaries:

```
0
32
64
96
128
160
192
224
```

`150` falls in:

```
128 - 159
```

Therefore:

```
Network:
192.168.10.128
```

Broadcast:

```
192.168.10.159
```

Usable:

```
192.168.10.129
        ↓
192.168.10.158
```

---

19. Connect This to Day 7

This is where subnetting becomes practically important.

Suppose:

```
PC-A
IP = 192.168.1.10/24
```

and:

```
PC-B
IP = 192.168.1.20/24
```

PC-A calculates:

```
Network A = 192.168.1.0
Network B = 192.168.1.0
```

Same network.

Therefore:

```
PC-A
  |
  | ARP for PC-B
  ▼
MAC-B
  |
  ▼
Switch
  |
  ▼
PC-B
```

No router is required for the local delivery.

---

20. Different Network

Now:

```
PC-A
192.168.1.10/24
```

PC-B:

```
192.168.2.20/24
```

PC-A calculates:

```
PC-A network = 192.168.1.0
PC-B network = 192.168.2.0
```

Different networks.

Therefore PC-A sends the frame toward its **default gateway**.

```
PC-A
192.168.1.10
      |
      | Destination IP:
      | 192.168.2.20
      ▼
Default Gateway
192.168.1.1
      |
      ▼
Router
      |
      ▼
192.168.2.0 network
      |
      ▼
PC-B
192.168.2.20
```

This is why subnetting matters to routing.

---

21. The Router's Decision

Suppose router has:

```
Interface 1:
192.168.1.1/24

Interface 2:
192.168.2.1/24
```

PC-A:

```
192.168.1.10
```

Destination:

```
192.168.2.20
```

Router receives the packet.

It checks:

```
Destination IP
       ↓
```

```
Routing table
      ↓
192.168.2.0/24
      ↓
Interface 2
```

Then it forwards the packet toward PC-B.

---

22. Important: IP and MAC Again

This is one of the concepts you should now be able to explain clearly.

For different networks:

IP addresses

Generally remain the same end-to-end:

```
Source IP      = PC-A
Destination IP = PC-B
```

MAC addresses

Change at every Layer-2 hop.

Example:

```
PC-A → Router

Src MAC = PC-A
Dst MAC = Router
```

Then:

```
Router → PC-B

Src MAC = Router
Dst MAC = PC-B
```

Visual:

```
          Layer 3
      ┌─────────────────┐
      │ Src IP = PC-A   │
      │ Dst IP = PC-B   │
      └─────────────────┘
           stays
             |
```

```
        ▼
┌─────────────────┐
│ Ethernet Frame  │
│ MAC-A → MAC-R   │
└─────────────────┘

        │
      Router
        │
        ▼

┌─────────────────┐
│ Ethernet Frame  │
│ MAC-R → MAC-B   │
└─────────────────┘
```

---

23. Private IPv4 Addresses

You should know these for interviews.

`10.0.0.0/8`

```
10.0.0.0 – 10.255.255.255
```

`172.16.0.0/12`

```
172.16.0.0 – 172.31.255.255
```

`192.168.0.0/16`

```
192.168.0.0 – 192.168.255.255
```

These are commonly used for private networks.

Example home network:

```
Router
192.168.1.1

Laptop
192.168.1.10

Phone
192.168.1.11

TV
192.168.1.20
```

---

24. Public vs Private IP

```
Private IP
    ↓
Used inside private networks
    ↓
Example:
192.168.1.10
```

```
Public IP
    ↓
Globally routable address
    ↓
Used for Internet communication
```

A home network commonly looks conceptually like:

```
Laptop
192.168.1.10
     |
     ▼
Home Router
Private IP: 192.168.1.1
Public IP:  ISP-assigned
     |
     ▼
Internet
```

NAT, which we'll cover later, allows many private hosts to share public connectivity.

---

25. Special IPv4 Addresses You Should Know

Address/Range	Meaning
`0.0.0.0`	Unspecified / context-dependent
`127.0.0.1`	IPv4 loopback
`127.0.0.0/8`	Loopback range
`169.254.0.0/16`	IPv4 link-local
`255.255.255.255`	Limited broadcast

The most important:

```
127.0.0.1
     ↓
localhost / loopback
```

Example:

```
Application
    |
    | connect(127.0.0.1, 8080)
    ▼
Same machine
```

The traffic doesn't need to travel through the physical network like normal remote traffic.

---

26. Linux Commands for Day 8

Since your interview target involves Linux, know these.

View IP addresses

```
ip addr
```

or:

```
ip a
```

View routing table

```
ip route
```

View interface information

```
ip link
```

Example:

```
$ ip route

default via 192.168.1.1 dev eth0
192.168.1.0/24 dev eth0
```

This tells you:

```
192.168.1.0/24
      ↓
directly connected network

default
      ↓
send other destinations to gateway
```

---

27. Very Important Routing Concept

Suppose:

```
Destination:
192.168.1.50
```

Routing table:

```
192.168.1.0/24 dev eth0
default via 192.168.1.1
```

Which route is selected?

```
192.168.1.0/24
```

Why?

Because it is a **more specific route** than the default route.

This leads to an important routing rule:

> **Longest-prefix match**

---

28. Longest Prefix Match

Suppose routing table contains:

```
10.0.0.0/8
10.10.0.0/16
10.10.20.0/24
default
```

Destination:

```
10.10.20.50
```

Matches:

```
10.0.0.0/8
10.10.0.0/16
10.10.20.0/24
```

Which one wins?

```
10.10.20.0/24
```

because `/24` is the longest/more-specific matching prefix.

Visual:

```
/8
 └──── matches

/16
 └──────── matches

/24
 └────────────── matches
                   ↑
                 BEST
```

This is fundamental to router forwarding.

---

29. Day 8 Complete Mental Model

```
                IPv4 Address
                     |
                     ▼
             32-bit address
                     |
                     ▼
               Subnet Mask
                     |
```

```
                         ▼
             ┌───────────┴───────────┐
             │                       │
             ▼                       ▼
       Network Part             Host Part
             │                       │
             ▼                       ▼
     Network Address          Host Address
             │
             ▼
     Routing Decision
             │
             ▼
     Same Network?
         │         │
       YES        NO
         │         │
         ▼         ▼
        ARP      Default Gateway
         │         │
         ▼         ▼
      Switch      Router
```

---

30. Day 8 Interview Questions

Before moving to Day 9, you should be able to answer these:

Basic

1. What is an IPv4 address?
2. How many bits are in IPv4?
3. What is an octet?
4. What is a subnet mask?
5. What does `/24` mean?
6. What is CIDR?

Subnetting

7. What is the network address?
8. What is the broadcast address?
9. How many usable hosts are in `/24` ?
10. How many usable hosts are in `/26` ?
11. How many `/26` subnets can be created from `/24` ?
12. Find the network address of `192.168.1.70/26` .

13. Find the broadcast address of `192.168.1.70/26`.
14. Find the network address of `192.168.10.150/27`.

Routing

15. How does a host determine whether the destination is on the same network?
16. What happens when the destination is on a different network?
17. What is a default gateway?
18. What is longest-prefix match?
19. Why does a router need a subnet mask/prefix?
20. What changes when a packet crosses a router: IP address or MAC address?

⭐ Day 8 Interview Summary

Memorize this chain:

```
IPv4
 ↓
32 bits
 ↓
Network + Host
 ↓
Subnet Mask / CIDR
 ↓
Network Address
 ↓
Broadcast Address
 ↓
Usable Host Range
 ↓
Same Network?
 ├── YES → ARP → Destination MAC → Switch
 │
 └── NO  → ARP → Gateway MAC → Router
                       ↓
                   Routing Table
                       ↓
                  Next Network
```

The 5 formulas/concepts to remember

```
1. IPv4 = 32 bits

2. Host bits = 32 - prefix length
```

```
3. Total addresses = 2^(host bits)

4. Traditional usable hosts = 2^(host bits) - 2

5. Router forwarding = Longest Prefix Match
```

Day 8 is complete when you can calculate network address, broadcast address, host range, and determine whether PC-A sends directly to PC-B or to the default gateway.

Next in the fixed schedule: Day 9 — Routing Fundamentals + Routing Tables.

Today 7:54 AM

18. Another Example Find the network and broadcast for: /27: Block size: Subnet boundaries: 150 falls in: Therefore: Broadcast: Usable:

explain clearlly

Absolutely. Let's slow down and understand **why** `192.168.10.150/27` **gives network** `192.168.10.128` **and broadcast** `192.168.10.159` . This is a very important subnetting calculation for interviews.

Example: `192.168.10.150/27`

We need to find:

1. Network address
2. Broadcast address
3. First usable IP
4. Last usable IP

Step 1 — Understand `/27`

IPv4 has **32 bits**.

```
32 bits total
```

`/27` means:

```
27 bits → Network
 5 bits → Host
```

Because:

```
32 - 27 = 5
```

So:

```
192.168.10.150/27

┌─────────────────────────────┬──────┐
│       Network = 27 bits     │Host  │
│                             │5 bit │
└─────────────────────────────┴──────┘
```

Step 2 — Find the subnet mask

`/27` means 27 `1` s followed by 5 `0` s.

```
11111111.11111111.11111111.11100000
```

Convert each octet:

```
11111111 = 255
11111111 = 255
11111111 = 255
11100000 = 224
```

Therefore:

```
/27 = 255.255.255.224
```

Step 3 — Why do we calculate `256 - 224` ?

The interesting part is the **last octet**.

Our mask is:

```
255.255.255.224
             ↑
       important octet
```

Calculate:

```
256 - 224 = 32
```

This `32` is called the **block size**.

It tells us:

> Each `/27` subnet contains 32 IP addresses.

Step 4 — Create the subnet boundaries

If every subnet has **32 addresses**, start from `0` and keep adding 32:

```
0
0 + 32 = 32
32 + 32 = 64
64 + 32 = 96
96 + 32 = 128
128 + 32 = 160
160 + 32 = 192
192 + 32 = 224
```

So the subnet boundaries are:

```
0
32
64
96
128
160
192
224
```

These are the **network addresses**.

Therefore:

```
192.168.10.0/27
192.168.10.32/27
192.168.10.64/27
192.168.10.96/27
192.168.10.128/27
192.168.10.160/27
192.168.10.192/27
192.168.10.224/27
```

Step 5 — Where does `150` belong?

Our IP is:

```
192.168.10.150
```

We only need to look at the last octet:

```
150
```

Look at our ranges:

```
0   ——— 31
32  ——— 63
64  ——— 95
96  ——— 127
128 ——— 159   ← 150 is here
160 ——— 191
192 ——— 223
224 ——— 255
```

`150` is between:

```
128 and 159
```

Therefore, the subnet containing `150` starts at:

```
128
```

So:

Network Address = `192.168.10.128`

Step 6 — Find the broadcast address

Our subnet is:

```
128 ——— 159
```

The **last address** in that subnet is the broadcast address.

Therefore:

```
Broadcast = 192.168.10.159
```

That's it.

Step 7 — Find usable host addresses

Remember:

```
Network address
       ↓
cannot normally be assigned to a host

Broadcast address
       ↓
cannot normally be assigned to a host
```

Our subnet:

```
192.168.10.128
        ↓
     Network

192.168.10.129
        ↓
     First host

...

192.168.10.158
        ↓
     Last host

192.168.10.159
        ↓
    Broadcast
```

Therefore:

```
Network:
192.168.10.128

First usable:
192.168.10.129

Last usable:
192.168.10.158
```

```
Broadcast:
```

Visualize the Complete Subnet

```
              192.168.10.0/24
                     │
         Divide into /27 subnets
                     │
                     ▼

┌─────────────────────────────────────┐
│ 192.168.10.0   → 31                 │
│ Network          Broadcast          │
└─────────────────────────────────────┘

┌─────────────────────────────────────┐
│ 192.168.10.32  → 63                 │
│ Network          Broadcast          │
└─────────────────────────────────────┘

┌─────────────────────────────────────┐
│ 192.168.10.64  → 95                 │
│ Network          Broadcast          │
└─────────────────────────────────────┘

┌─────────────────────────────────────┐
│ 192.168.10.96  → 127                │
│ Network          Broadcast          │
└─────────────────────────────────────┘

┌─────────────────────────────────────┐
│ 192.168.10.128 → 159                │
│ Network                          ↑    │
│     ↑                           150   │
│     │                               │
│     └── Our subnet                  │
└─────────────────────────────────────┘

┌─────────────────────────────────────┐
│ 192.168.10.160 → 191                │
└─────────────────────────────────────┘

┌─────────────────────────────────────┐
│ 192.168.10.192 → 223                │
└─────────────────────────────────────┘

┌─────────────────────────────────────┐
│ 192.168.10.224 → 255                │
└─────────────────────────────────────┘
```

The Most Important Trick

For subnetting questions like:

```
192.168.10.150/27
```

you don't need to convert everything to binary every time.

Use this **4-step interview method**:

① Find mask

```
/27
 ↓
255.255.255.224
```

② Calculate block size

```
256 - 224 = 32
```

③ Find the range containing 150

```
0–31
32–63
64–95
96–127
128–159  ← 150
160–191
192–223
224–255
```

④ Answer

```
Network   = 192.168.10.128
Broadcast = 192.168.10.159

Usable:
192.168.10.129 – 192.168.10.158
```

Why is `160` NOT the network for `150`?

This is a common confusion.

You might think:

```
150 is closer to 160
```

But that's not how subnet boundaries work.

The boundaries are fixed:

```
0 → 32 → 64 → 96 → 128 → 160 → 192 → 224
```

`150` comes **after 128 but before 160**:

```
128 ─────────────── 150 ─────────────── 159
 ↑                   ↑                   ↑
Network             Host             Broadcast
```

Therefore:

```
Network = 128
```

not `160`.

One More Example

Try:

```
192.168.10.200/27
```

We already know `/27` gives blocks of `32`:

```
0–31
32–63
64–95
96–127
128–159
160–191
192–223  ← 200 is here
224–255
```

Therefore:

```
Network   = 192.168.10.192
Broadcast = 192.168.10.223
```

```
First      = 192.168.10.193
Last       = 192.168.10.222
```

⭐ Remember This Pattern

```
              /27
               |
               ▼
        Mask = 224
               |
               ▼
       256 - 224 = 32
               |
               ▼
      Block size = 32
               |
               ▼
0, 32, 64, 96, 128, 160, 192, 224
               |
               ▼
       Find where IP falls
               |
               ▼
       Network = block start
               |
               ▼
     Broadcast = next block - 1
```

So for:

```
192.168.10.150/27
```

the answer is:

```
┌─────────────────────────────┐
| Network   : 192.168.10.128 |
| First     : 192.168.10.129 |
| Last      : 192.168.10.158 |
| Broadcast : 192.168.10.159 |
└─────────────────────────────┘
```

This block-size method is the one I recommend you use first in interviews. Once this is comfortable, binary calculation becomes much easier to understand rather than something you have to memorize.